

Assignment 19 (ELEC 341 L19_RelativeStability)

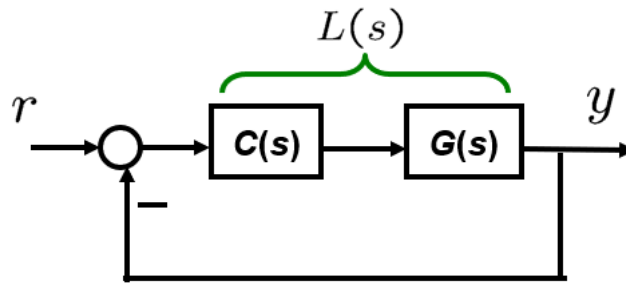
Problem 1:

For the system shown in the following figure, where

$$G(s) = \frac{1}{(s+5)(s+20)(s+50)}$$

Do the following:

- Draw the Bode log-magnitude and phase plots.
- Evaluate gain margin, phase margin, gain cross-over frequency, and phase cross-over frequency analytically and show them in your Bode plots for $C(s) = 10,000$.



$$a). \quad L(s) = \frac{10K}{(s+5)(s+20)(s+50)} \quad \rightarrow \quad \frac{10K}{s^3 + 75s^2 + 1350s + 5000} = \frac{10K}{(5000 - 75s^2) + j(1350s - s^3)}$$

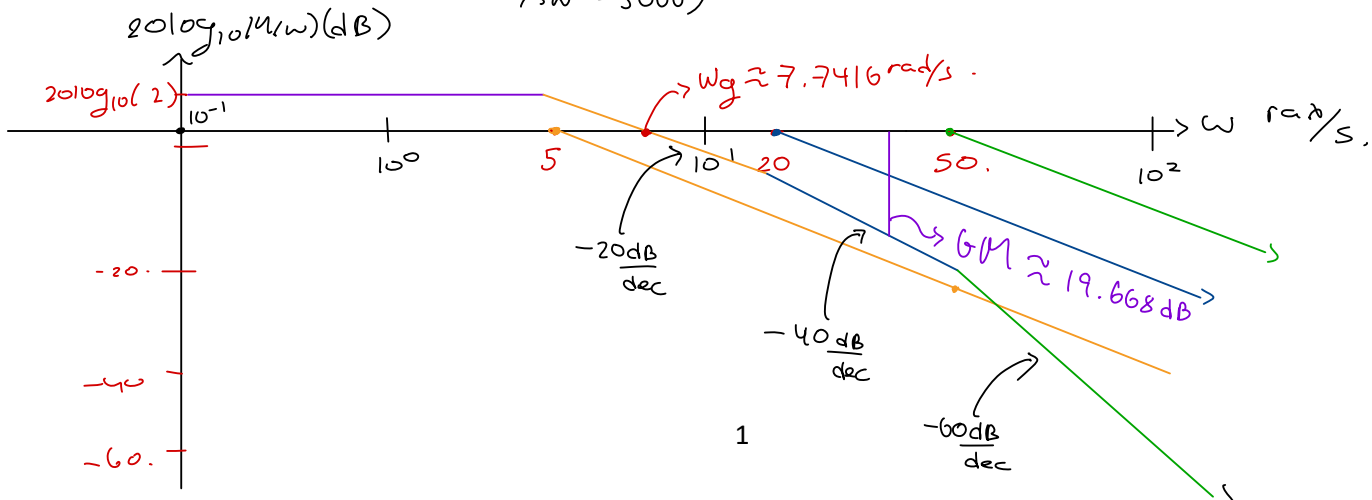
$$L(j\omega) = \frac{10K}{(j\omega+5)(j\omega+20)(j\omega+50)} = \frac{10K}{5(20)(50)(1+\frac{j\omega}{5})(1+\frac{j\omega}{20})(1+\frac{j\omega}{50})}$$

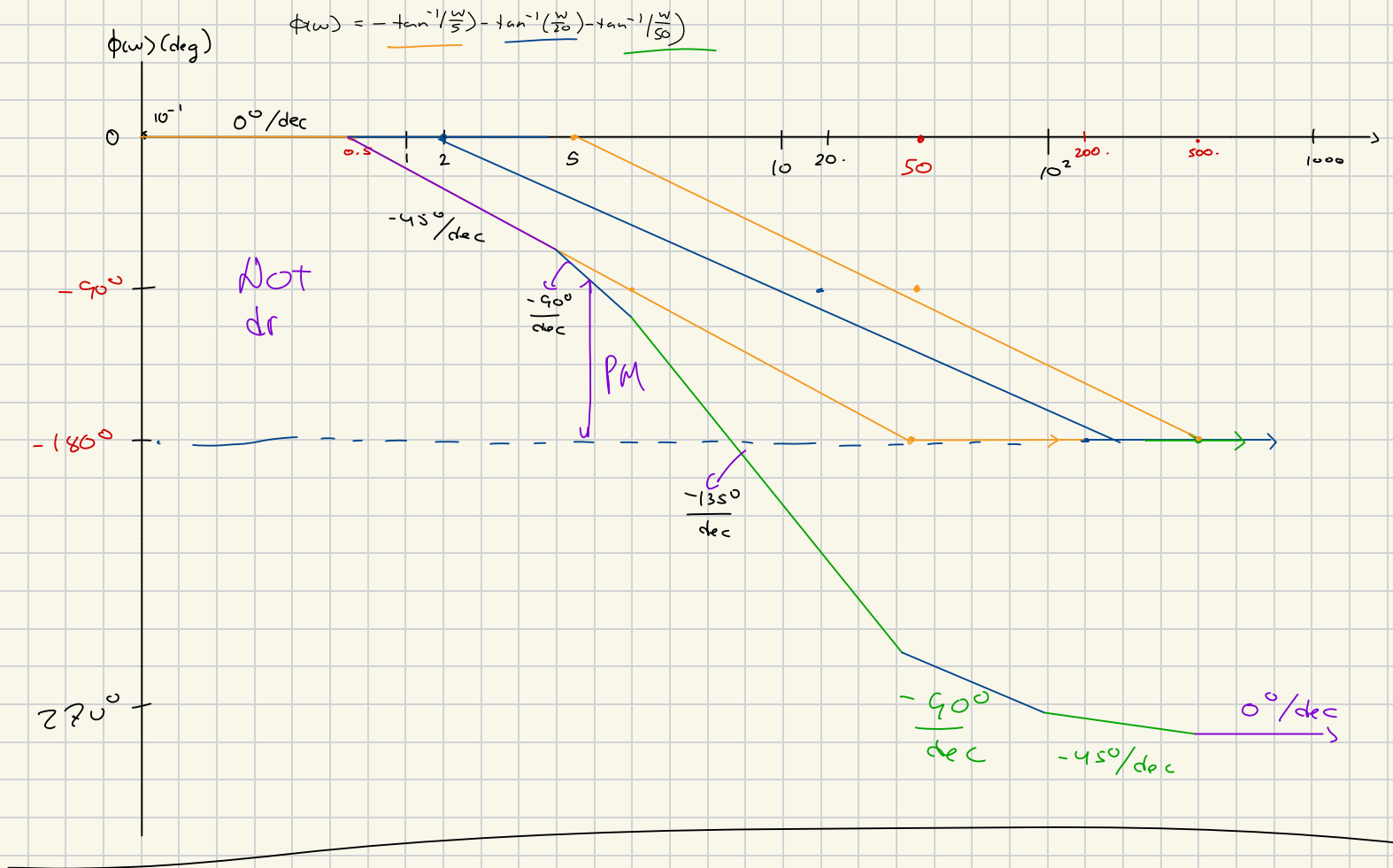
$$M(\omega) = \frac{2}{\sqrt{\omega^2+25}\sqrt{\omega^2+400}\sqrt{\omega^2+2500}} = \frac{2}{(1+\frac{j\omega}{5})(1+\frac{j\omega}{20})(1+\frac{j\omega}{50})}$$

$$20\log_{10} M(\omega) = 20\log_{10}(2) - 20\log_{10}\left(1+\frac{\omega}{5}\right) - 20\log_{10}\left(1+\frac{\omega}{20}\right) - 20\log_{10}\left(1+\frac{\omega}{50}\right)$$

$$\phi(\omega) = -\tan^{-1}\left(\frac{1350\omega - \omega^3}{5000 - 75\omega^2}\right)$$

$$\phi(\omega) = -\tan^{-1}\left(\frac{\omega^3 - 1350\omega}{75\omega^2 - 5000}\right) \quad \text{or} \quad \phi(\omega) = -\tan^{-1}\left(\frac{\omega}{5}\right) - \tan^{-1}\left(\frac{\omega}{20}\right) - \tan^{-1}\left(\frac{\omega}{50}\right)$$





b). $GM = 20 \log_{10} \left(\frac{1}{|L(j\omega_p)|} \right)$, $\angle L(j\omega_p) = -180^\circ$ $L(s) = c(s)G(s)$

$PM = \angle L(j\omega_g) + 180^\circ$, $|L(j\omega_g)| = 1$.

$L(j\omega) = \frac{10K}{\sqrt{\omega^2 + 25} \sqrt{\omega^2 + 400} \sqrt{\omega^2 + 2500}}$

$\angle L(\omega_p) = \phi(\omega_p) = -\tan^{-1} \left(\frac{\omega_p^3 - 1350\omega_p}{75\omega_p^2 - 5000} \right) = -180^\circ$ or $-\tan^{-1}(\frac{\omega}{5}) - \tan^{-1}(\frac{\omega}{20}) - \tan^{-1}(\frac{\omega}{50})$

$\omega_p^3 - 1350\omega_p = 0$

$\omega_p(\omega_p^2 - 1350) = 0$

$\omega_p = 0, 15\sqrt{6}$

$\omega_p = 36.742346 \text{ rad/s}$

(occurs when $\phi(\omega) = -180^\circ$)

$L(j\omega_p) = \frac{10000}{\sqrt{\omega_p^2 + 25} \sqrt{\omega_p^2 + 400} \sqrt{\omega_p^2 + 2500}}$

$L(j\omega_p) = 0.1038961$

$GM = 20 \log_{10} \left(\frac{1}{|L(j\omega_p)|} \right)$

$GM = 19.668015 \text{ dB}$

↳ the vertical distance when ω_p occurs.

Find PM and ω_g .

$|L(j\omega_g)| = 1$

$10K = 10^4$

1

$\frac{10K}{\sqrt{\omega_g^2 + 25} \sqrt{\omega_g^2 + 400} \sqrt{\omega_g^2 + 2500}} = 1$

$PM = \angle L(j\omega_g) + 180^\circ$

$\angle L(j\omega_g) = \phi(\omega_g) = -\left(\tan^{-1}(\frac{\omega_g}{5}) + \tan^{-1}(\frac{\omega_g}{20}) + \tan^{-1}(\frac{\omega_g}{50}) \right)$

$\angle L(j\omega_g) = -87.1049^\circ$

$PM = 92.8951^\circ$

↳ amount of deg above 180° line where ω_g occurs.

$(\omega_g^2 + 25)(\omega_g^2 + 400)(\omega_g^2 + 2500) = 10^8$

used graphing calculator. $\omega_g = 7.7416 \text{ rad/s}$ occurs at intersection of graph with ω axis.